



Solve with us 3.2 - Not Graded



This assignment will not be graded and is only for practice.

Key points: 1

- Definition of linear dependence: A set of vectors $v_1, v_2, \dots, v_n \in V$ is said to be linearly dependent if there exist scalars $a_1, a_2, \dots, a_n \in R$, not all zero such that $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$.
- To check the vectors $v_1, v_2, \dots, v_n \in R^m$ (with usual addition and scalar multiplication) are linearly dependent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has infinitely many solutions, where the j^{th} column of A is v_j .
- Definition of linear independence: A set of vectors $v_1, v_2, \dots, v_n \in V$ is said to be linearly independent if $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$ implies that $a_i = 0$ for $i = 1, 2, \dots, n$.
- To check the vectors $v_1, v_2, \dots, v_n \in R^m$ (with usual addition and scalar multiplication) are linearly independent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has only the trivial solution (i.e., $x = 0$), where the j^{th} column of A is v_j .
- If S is a set containing zero vector, then the set S is a linearly dependent set.
- Let S be a set of vectors from a vector space V . If an element of S is a scalar multiple of another element from the set S , then S is a linearly dependent set.

1 point

Vectors $(2, 1)$ and $(6, 3)$ from R^2 are

- ☐ linearly independent.
☐ linearly dependent.
☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

1 point

Suppose the set $\{(3, 7, 4), v\}$ is a linearly dependent set in the vector space R^3 with usual addition and scalar multiplication. Which of the following vectors are possible candidates for v ?

☐

$(0, 0, 0)$

☐

$(5, \frac{35}{3}, \frac{20}{3})$

☐

$(1, 0, 0)$

☐

$(6, 14, -8)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, 0)$

$(5, \frac{35}{3}, \frac{20}{3})$

1 point

The set of vectors $\{(1, 1, -1), (-1, 1, 1), \text{ and } (0, \frac{1}{2}, 0)\}$ in R^3 is

- ☐ linearly independent.



☐ linearly dependent.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

1 point

$M_{2 \times 2}(R)$ denotes the vector space consisting of real square matrices of order 2 with usual matrix addition and scalar multiplication. Consider the following matrices:

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Which of the following options is(are) true?

[Hint: Form a system of linear equations using matrix equation $x \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + y \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$]

☐

$\{M_1, M_2\}$ is a linearly independent set.

☐

$\{M_1, M_2, M_3\}$ is a linearly dependent set.

☐

$\{M_2, M_3\}$ is a linearly dependent set.

☐

$\{M_1, M_3\}$ is a linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2\}$ is a linearly independent set.

$\{M_1, M_3\}$ is a linearly independent set.

Key points: 2

- Given two vectors (x_1, y_1) , and $(x_2, y_2) \in R^2$ construct $A = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}$. Then these vectors are linearly dependent if and only if $\det(A) = 0$ and linearly independent if and only if $\det(A) \neq 0$.

- Given three vectors (x_1, y_1, z_1) , (x_2, y_2, z_2) and $(x_3, y_3, z_3) \in R^3$ construct $A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$. Then these vectors

are linearly dependent if and only if $\det(A) = 0$ and linearly independent if and only if $\det(A) \neq 0$.

are linearly dependent if and only if $\det(A) = 0$ and linearly independent if and only if $\det(A) \neq 0$.

1 point

Vectors $(2, 3)$ and $(4, 5)$ from R^2 are

[Hint: Form the matrix using the vectors and calculate the determinant of the matrix.]

- ☐ linearly independent.
- ☐ linearly dependent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly independent.

1 point

Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\} \subset \mathbb{R}^2$. Choose the set of correct options.

[Hint:Hint: Form the matrix using the vectors as the columns for each option and calculate the determinant of the matrix.]

☐ $\{(1, 2), (-1, 2)\}$ is a linearly dependent set.

☐ $\{(-1, 2), (-3, 1)\}$ is a linearly independent set.

☐ $\{(-3, 1), (-1, 2)\}$ is a linearly independent set.

☐ $\{(3, 1), (-3, 1)\}$ is a linearly dependent set.

☐ $\{(3, 1), (1, 2)\}$ is a linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-1, 2), (-3, 1)\}$ is a linearly independent set.

$\{(-3, 1), (-1, 2)\}$ is a linearly independent set.

$\{(3, 1), (1, 2)\}$ is a linearly independent set.

1 point

The set of vectors $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is

[Hint:Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- ☐ linearly dependent.
- ☐ linearly independent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly independent.

1 point

Suppose $\{(2, 3, 0), (0, 1, 0), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible candidates for v ?

☐ $(2, 3, 1)$

☐ $(1, 2, 0)$

☐ $(1, 3, 0)$

 $(0, 1, 1)(0, 1, 1)$  $(\pi, e, 0)(\pi, e, 0)$ **No, the answer is incorrect.****Score: 0****Accepted Answers:** $(1, 2, 0)(1, 2, 0)$ $(1, 3, 0)(1, 3, 0)$ $(\pi, e, 0)(\pi, e, 0)$ *1 point*

Consider three vectors $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$ $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$ in the vector space R^3 R^3 , with usual addition and scalar multiplication. Which of the following option is true?

[Hint:Hint: Find out the vectors of the given sets in the options and form the matrices using the vectors as the columns and calculate the determinant of the matrices.]

 $\{v_1, v_2, v_3\}$ $\{v_1, v_2, v_3\}$ is a linearly dependent set. $\{v_1 + v_2, v_1 + v_3, v_3\}$ $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set. $\{v_2 - v_1, v_1, v_2\}$ $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.**No, the answer is incorrect.****Score: 0****Accepted Answers:** $\{v_2 - v_1, v_1, v_2\}$ $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

Key points: 3Key points: 3

- Any set of n vectors in R^2 with $n \geq 3$ is linearly dependent.
- Any set of n vectors in R^3 with $n \geq 4$ is linearly dependent.
- Any set of $n + 1$ vectors in R^n is linearly dependent.
- Any set containing a linearly dependent set is linearly dependent.
- If a set is linearly independent, then every subset of it, is linearly independent.

*1 point*The subset $S = \{(3, 1), (1, 2), (-3, 1)\}$ $S = \{(3, 1), (1, 2), (-3, 1)\}$ of R^2 R^2 is

[Hint: Hint: Any set of n vectors in R^2 with $n \geq 3$ is linearly dependent.]



linearly dependent.



linearly independent.



None of the above.

No, the answer is incorrect.**Score: 0****Accepted Answers:**

linearly dependent.

1 point

Set $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$ $S = \{(1, 1, -1), (-1, 1, 1), (0, 21, 0), (0, 1, -2), (1, 0, -2)\}$ subset of R^3 R^3 is

[Hint:Hint: Any set of n vectors in R^3 with $n \geq 4$ are linearly dependent.]

- ☐ linearly dependent.
☐ linearly independent.
☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

1 point

Let S be a linearly dependent subset of R^3 . Which of the following options are true?

☐

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

☐

$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.

☐

$S \setminus \{v\}$ must be linearly dependent for any $v \in S$.

☐

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

Check Answers

Your score is: 0/12